2026 BROCK AI DAY · 3-MINUTE RESEARCH PRESENTATION

Are language models converging on shared structures?

Do AI models built by different teams, from different text, organize ideas the same way?

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BACKGROUND: THREE IDEAS

How to look inside a language model

1

Mechanistic interpretability

Study a model by reading its internal activity, not just its answers. That needs the model's weights, so we use open models.

2

Ideas are directions

A model stores each idea it learns, like *negation* or *credibility*, as a direction inside it. The Linear Representation Hypothesis.

3

One world, one map?

Models trained on different data may converge on the same picture of the world. The Platonic Representation Hypothesis.

Park, Choe & Veitch (2024), *The Linear Representation Hypothesis and the Geometry of Large Language Models*, ICML · Huh, Cheung, Wang & Isola (2024), *The Platonic Representation Hypothesis*, ICML.

PROBLEM

Do different models learn the same map of ideas?

Every model stores ideas in its own private coordinates, so you cannot lay one model's directions over another's.

But the **relationships** between ideas could still match: is credibility close to formality, far from sarcasm, in every model?

WHY IT MATTERS

If the map is shared, what we learn inside one model can be checked in others.

And it tells us whether models learn something about the world, or just quirks of their training.

METHOD

Find each idea, then compare the angles

1

Write pairs

The same passage twice, with and without the idea. 17 ideas, 250 pairs each.

2

Find the directions

Run both through 35 open models from 10 families. The difference is the idea's direction.

3

Compare angles

Inside each model, measure how close each idea is to every other. Angles can be compared across models; coordinates cannot.

4

Try to fail

Hide a whole model family. Can the other families' angles alone name its ideas?

Models include Llama, Mistral, Qwen, Gemma, Pythia, OPT and GPT-2, from 70 million to 72 billion parameters.

RESULTS

The other families name the hidden ideas

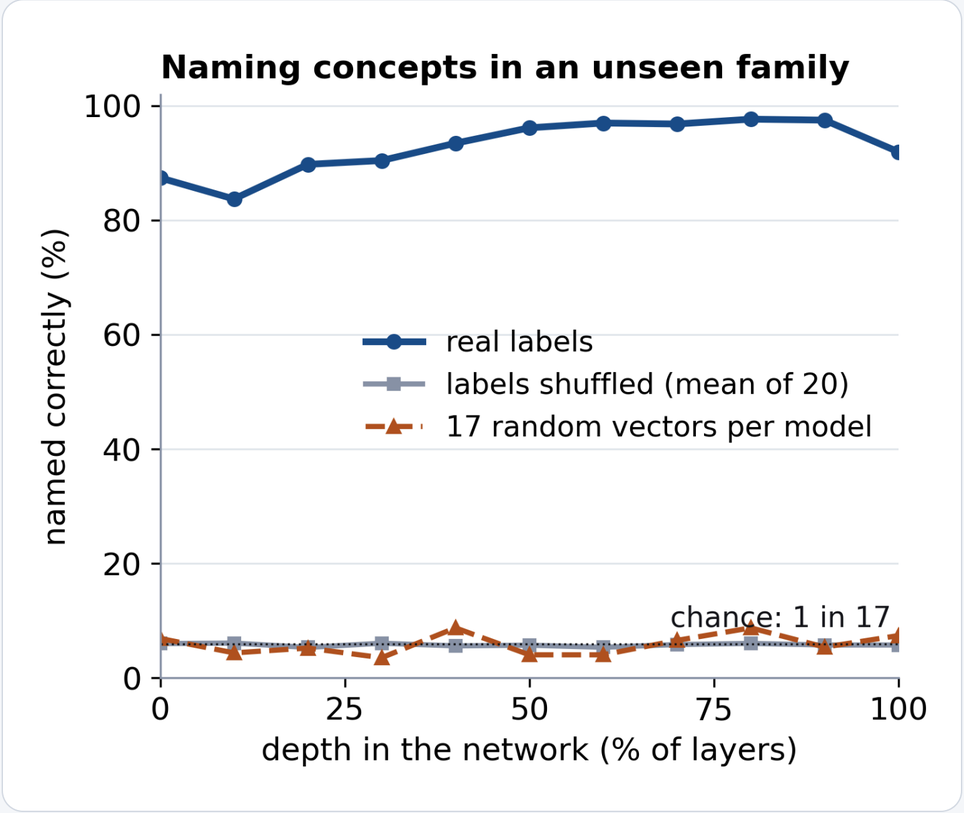
97%

of ideas named correctly in a hidden model family, in the middle layers

6%

with the labels shuffled, or with random directions: chance

One example: in 29 of 35 models, the idea most closely tied to credibility is formality.



CONCLUSION

Different models, one map of ideas

Language models built by different teams, from different text, organize ideas the same way. That supports the Platonic Representation Hypothesis, measured idea by idea.

Because the map is shared, findings from inside one model can be tested on others.

Is this the structure of language itself, or of the world it describes?

Come and see the poster. Code, data and reading list: github.com/jamesrahenry/ai-day-2026